ORF363/COS323 Convexity & Convex Optimization Part 1 (Lec4/5.pdf)

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Topics & Logistics

- Convex analysis
  - Convex sets & functions
  - Fundamental theorem of convex analysis
- **Examples** of convex sets & functions
- **Proofs** on blackboard & slides!

- **HW1 deadline now due Feb. 8, 11:59pm**
  - **Because of the new students**
- **My Office hours:**
  - **Fridays 4-6pm @ 123 Sherrerd**

General Constrained Optimization problem

- Recall the general constrained optimization problem

$$\min_{x \in R^n} \quad f(x)$$

$$\text{s.t.} \qquad x \in \Omega$$

$$f: R^n \to R$$

$$\Omega = \{x \in R^n \mid \; g_i(x) \leq 0 \,, i = 1,2, \dots m \quad and \quad h_j(x) = 0, j = 1,2, \dots, r \}$$

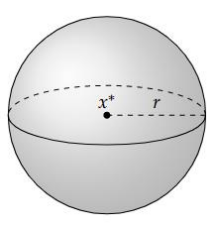
where: $g_i: R^n \to R$ and $h_j: R^n \to R$, for all i, j

- We make no assumptions on the type of the objective and constraint functions.
  - They can be convex, concave, or nonconvex, affine, quadratic, or nonlinear functions
- We may have (many) local minima and maxima and saddle points in general constrained optimization problems
- Local minima and global minima have a very similar definition in constrained optimization problems

Local/Global Optimality for General **Constrained Optimization**

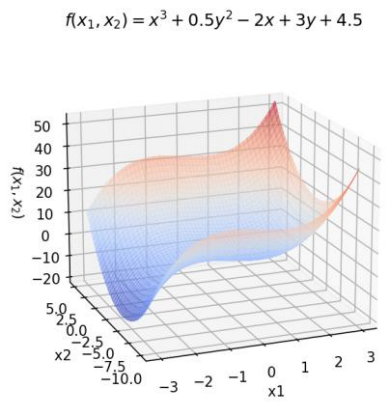
- x^* is a **local minimum** iff it is feasible (i.e., $x^* \in \Omega$) and $\exists\, r > 0$ such that
  - $f(x^*) \le f(x), \forall \boldsymbol{x} \in \boldsymbol{B}(\boldsymbol{x}^*, \boldsymbol{r}) \cap \boldsymbol{\Omega}$, where $\boxed{B(x^*, r) = \{x \in R^n \mid \|x - x^*\| \le r\}}$
- x^* is a **strict local minimum** iff it is feasible (i.e., $x^* \in \Omega$) and $\exists\, r > 0$ such that
  - $f(x^*) < f(x), \forall x \in B(x^*, r) \cap \Omega$ and $x \ne x^*$
- x^* is a **global minimum** iff it is feasible (i.e., $x^* \in \Omega$) and
  - $f(x^*) \le f(x), \forall \boldsymbol{x} \in \boldsymbol{\Omega}$
- x^* is a **strict global minimum** iff $x^* \in \Omega$ it is feasible (i.e., $x^* \in \Omega$) and
  - $f(x^*) < f(x), \forall x \in \Omega$ and $x \ne x^*$

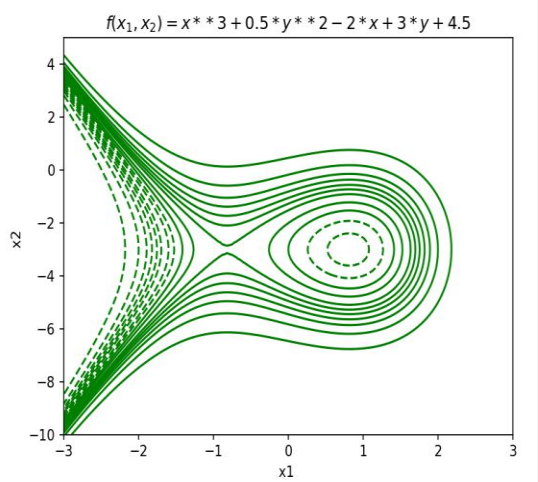
$B(x^*, r)$



Euclidean ball

$B(x^*, r)$ defines a neighborhood in the feasible region within which x^* is the local minimum

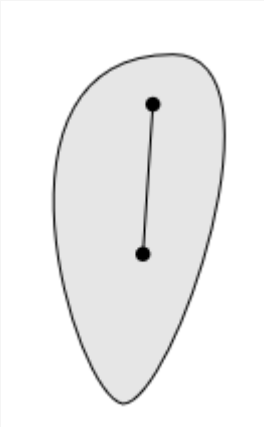


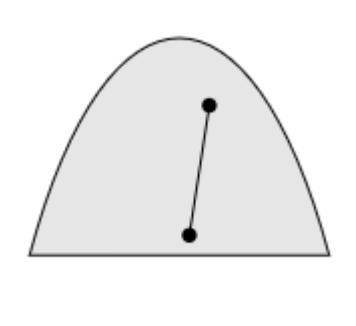


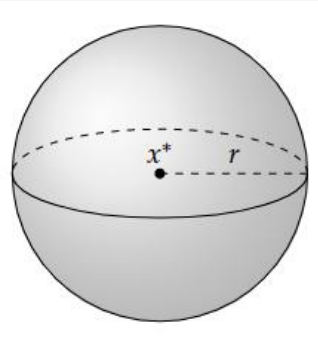
Convex analysis

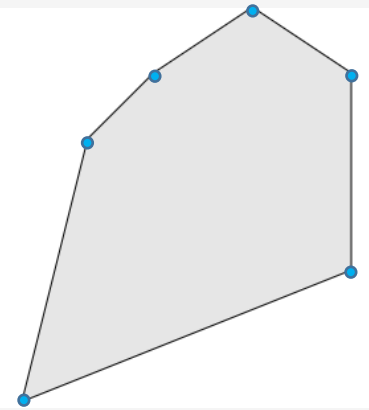
- $\Omega \subseteq R^n$ is **convex set** iff $\forall x, y \in \Omega, \lambda \in [0,1]$ the point $\lambda x + (1-\lambda)y \in \Omega$

- **Examples**:









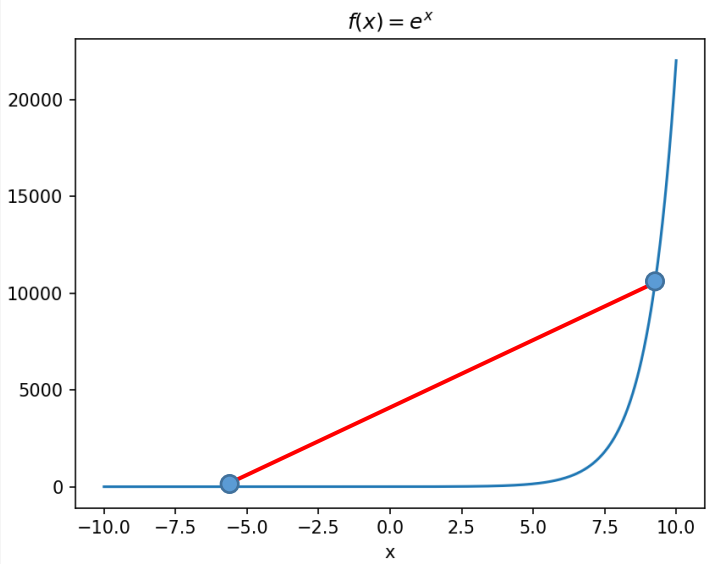
- **Exercise:** Prove that the Euclidean ball is a convex set!!

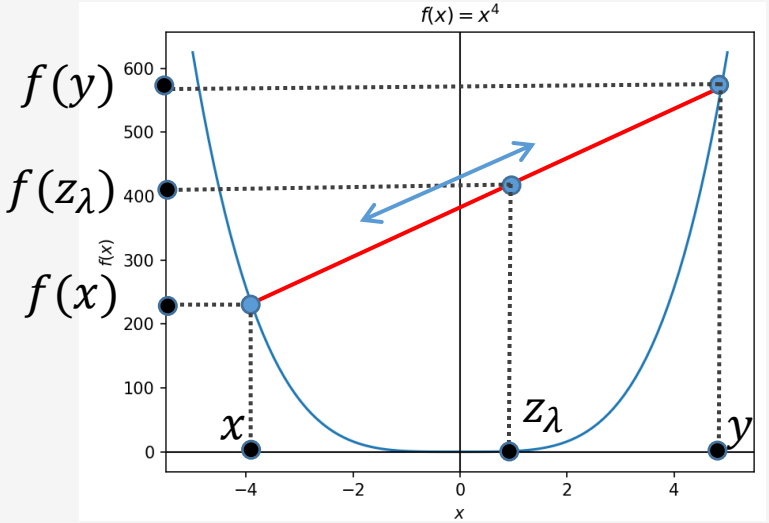
Convex analysis

- $f: R^n \to R$ **convex function** iff $\forall x, y \in dom(f), \lambda \in [0,1]$ we have

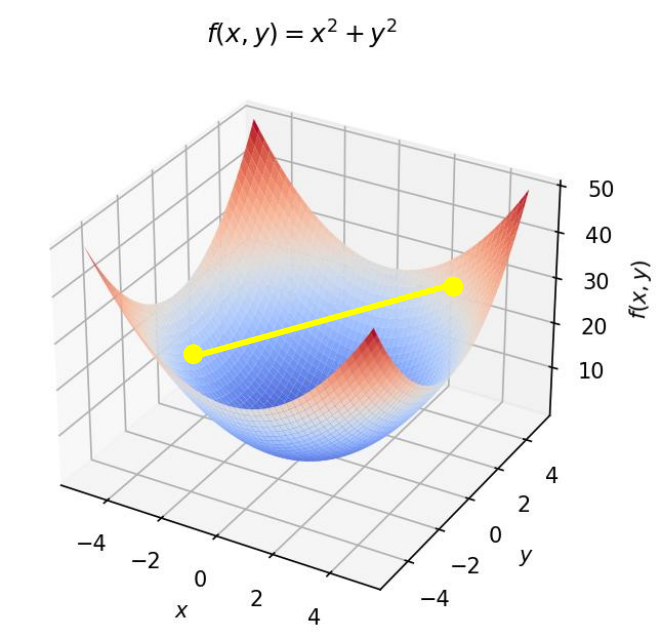
$$f(\lambda x + (1-\lambda)y) \leq \lambda f(x) + (1-\lambda) f(y)$$

- **Examples**:





$$z_\lambda = \lambda x + (1-\lambda) y$$



Convex analysis

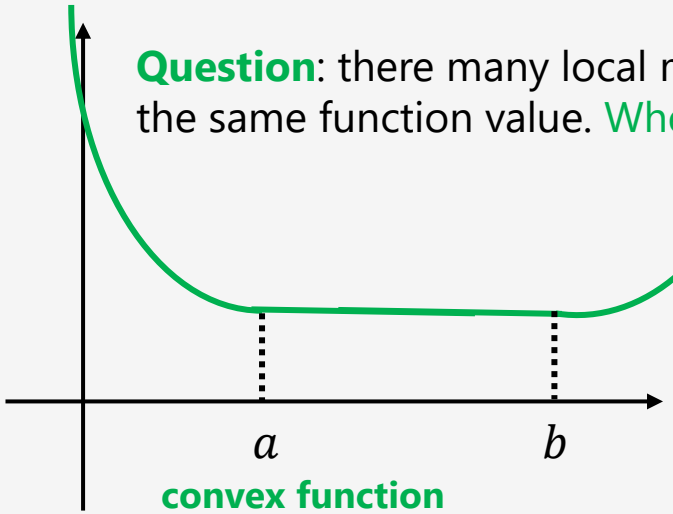
- $f: R^n \to R$ **strictly convex function** iff $\forall x, y \in R^n, \boxed{x \neq y}$, and $\forall \lambda \in \boxed{(0,1)}$

$$f(\lambda x + (1-\lambda)y) < \lambda f(x) + (1-\lambda) f(y)$$

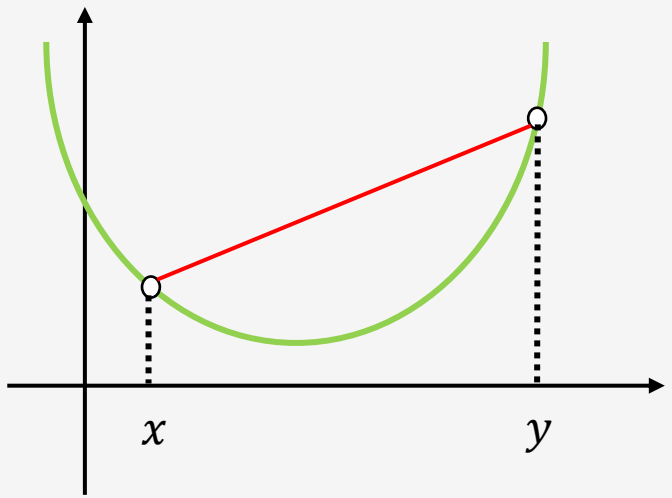
For convex functions we have $\leq$

$\forall x, y \in [a, b]$ and $\lambda \in [0,1]$ we have
$f(\lambda x + (1-\lambda)y) = \lambda f(x) + (1-\lambda) f(y)$

Question: there many local minima, with the same function value. Where??



convex function
(but not strictly convex)



strictly convex function

Convex analysis

- $f: R^n \to R$ is **concave function** iff $\forall x, y \in dom(f), \lambda \in [0,1]$ we have

$$f(\lambda x + (1-\lambda)y) \geq \lambda f(x) + (1-\lambda) f(y)$$

  - Or $-f$ is convex function

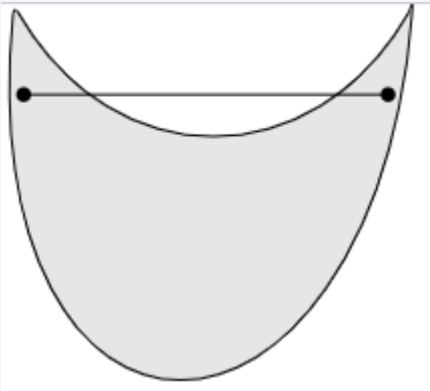
- $f: R^n \to R$ **strictly concave function** iff $\forall x, y \in R^n, x \neq y$, and $\forall \lambda \in (0,1)$

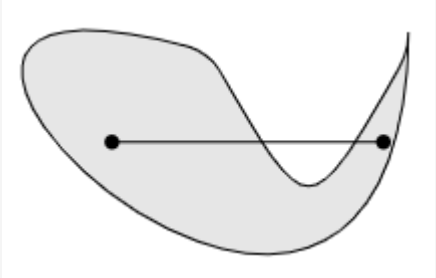
$$f(\lambda x + (1-\lambda)y) > \lambda f(x) + (1-\lambda) f(y)$$

  - Or $-f$ is strictly convex function

Convex analysis

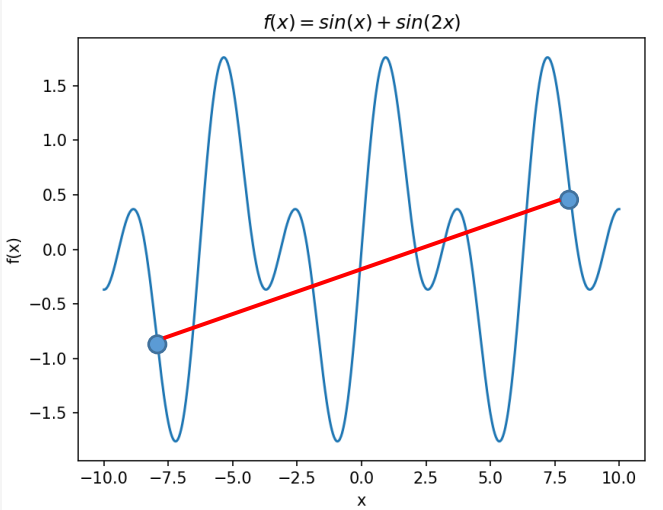
Nonconvex sets

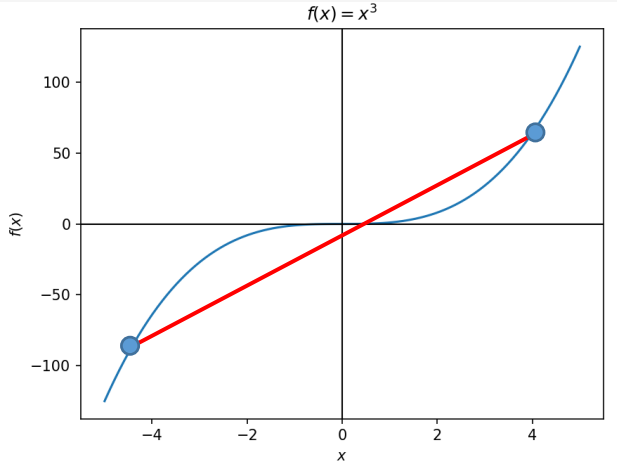


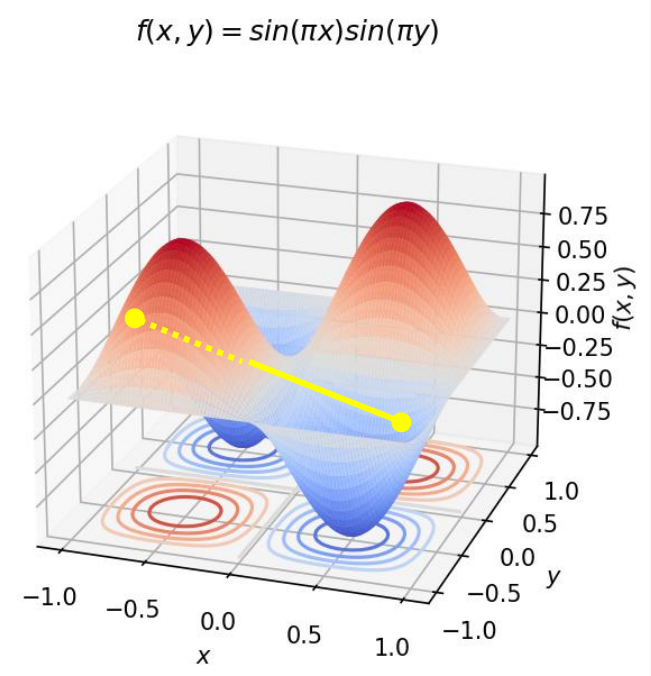


Question:
Find which points violate the Definition of the convex set

Nonconvex functions







Question:
Find which points violate the definition of convex function

The Convex Optimization problem

- A constrained optimization problem is called Convex Optimization problem iff:

$$\min_{x \in R^n} \; f(x)$$

$$\text{s.t.} \quad x \in \Omega$$

$$f: R^n \to R$$

$$\Omega = \{x \in R^n \mid \; g_i(x) \leq 0 \,, i = 1,2, \dots m \quad and \quad h_j(x) = 0, j = 1,2, \dots, r \}$$

where: $g_i: R^n \to R$ and $h_j: R^n \to R$, for all i, j

AND $f: R^n \to R$ is convex function,
AND all $g_i: R^n \to R$ are convex functions,
AND all $h_j: R^n \to R$, are affine functions.

$\Rightarrow \Omega$ is a convex set
WHY??
(see slide 17)

<u>Theorem:</u> Every local minimum of a convex optimization problem is also a global minimum

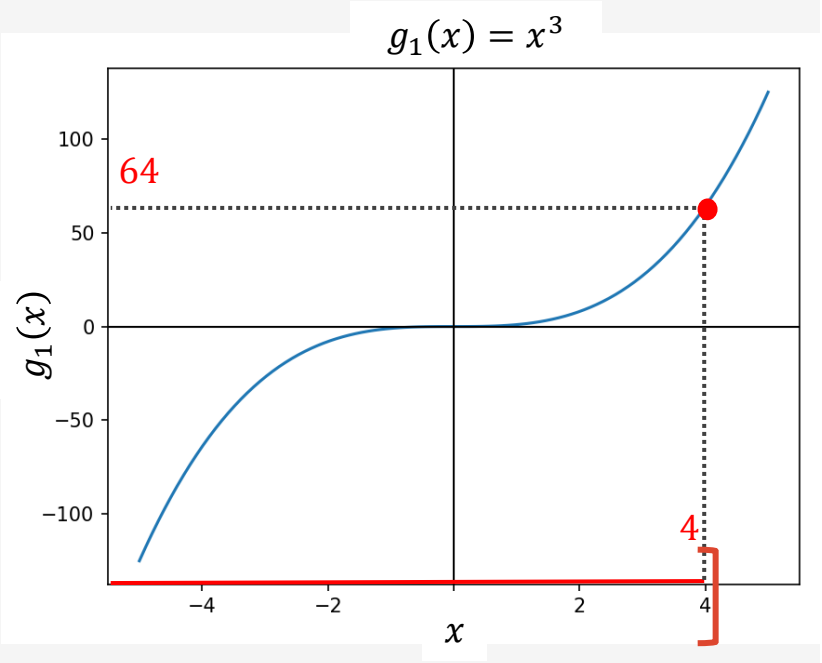
Proof: $\rightarrow$

The Convex Optimization problem

- **Proof**: we will use "**proof by contradiction**".
  - Suppose $\exists \hat{x} \in \Omega$ which is a **local minimum**, but **not a global minimum**.
  - Hence there is **another point** $y \in \Omega$ s.t. $f(y) \leq f(\hat{x})$ (1)
  - Consider a $\lambda \in [0,1)$. Then $\lambda\hat{x} + (1-\lambda)y \in \Omega$ (because Ω is convex set)
  - Also: $f(\lambda\hat{x} + (1-\lambda)y) \leq \lambda f(\hat{x}) + (1-\lambda)f(y)$ (because f is convex function)
  - Using equation (1) we have: $f(\lambda\hat{x} + (1-\lambda)y) \leq \lambda f(\hat{x}) + (1-\lambda)f(y) \overset{(1)}{\leq} \lambda f(\hat{x}) + (1-\lambda)f(\hat{x}) = f(\hat{x})$
  - Hence all points $\lambda\hat{x} + (1-\lambda)y$ connecting $\hat{x}$ and y are better than $\hat{x}$
  - But this is a **contradiction** since $\hat{x}$ is a local minimum in at least a neighbourhood. Hence **$\hat{x}$ is a global minimum**

The Convex Optimization problem

- **Remark**: For a Convex Optimization problem the feasible set Ω is a convex set, since all g_i are convex functions and all h_i are affine functions. (will prove it later)
  - **But the converse is not true (according to definition)**
  - **Example**: consider the feasible set $\Omega = \{x \in R \mid x^3 \leq 64\}$, where $g_1(x) = x^3$.
    - The set Ω is a convex set when we consider it by itself.
    - But if we place it within an optimization problem, with a convex objective function, it will not make it a Convex Optimization problem, because x^3 is not convex function.



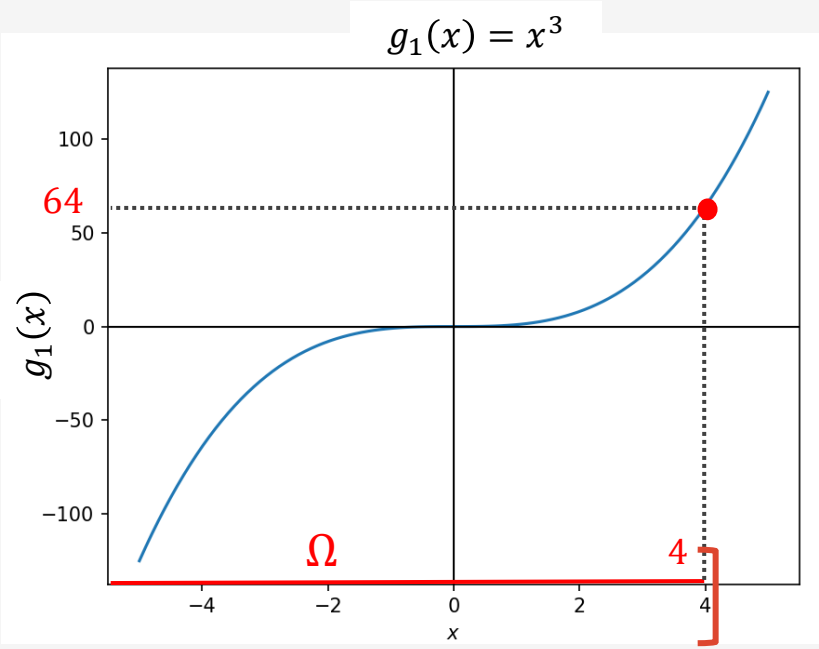
$$\Omega = \{x \in R \mid x^3 \leq 64\}$$

Is there any other constraint you can think of to **safely incorporate** within the convex optimization problem and still express the same constraint??

The Convex Optimization problem

- The **CVX solver** accepts Convex Optimization problems of the general form:
  - Objective must be convex func.
  - Constraints must have the form:
    - Convex func. ≤ Concave func.
    - Affine func. == Affine func.

Equivalently:
Convex func. ≤ 0
Affine func. == 0

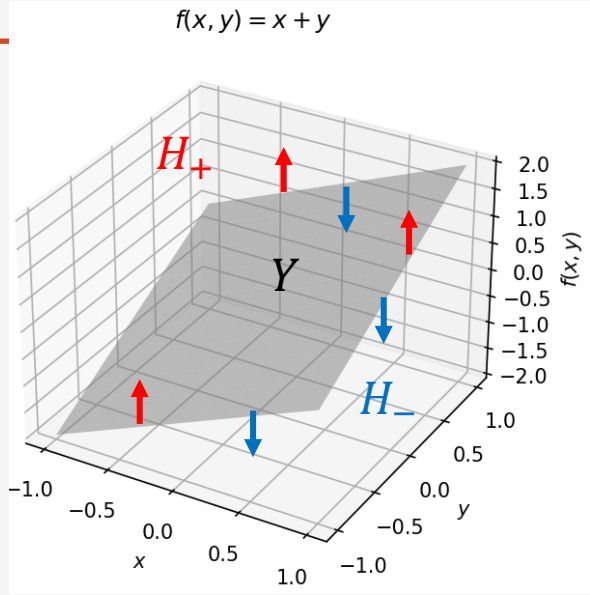


Bad modeling: $x^3 \leq 64$ not acceptable

Good modeling $x \leq 4$ acceptable

Frequently used convex sets

- **Hyperplane**: $Y = \{x \in R^n | a^T x = b\}$



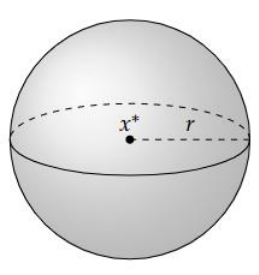
$\uparrow$ $a = \nabla(a^T x - b)$

$\downarrow$ $-a = -\nabla(a^T x - b)$

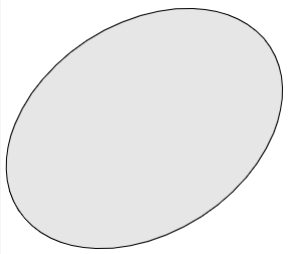
- **Halfspace**: $H_- = \{x \in R^n | a^T x \leq b\}$
  - $H_+ = \{x \in R^n | a^T x \geq b\}$

Exercise: Prove they are convex sets.

- **Euclidean ball**: $B(c, r) = \{x \in R^n | \left|\left|x - c\right|\right|^2 \leq r\}$



- **Ellipsoids**: $\mathrm{E}(c, r) = \{x \in R^n | \ (x - c)^T P \ (x - c) \leq r\}$, where $P \succ 0$



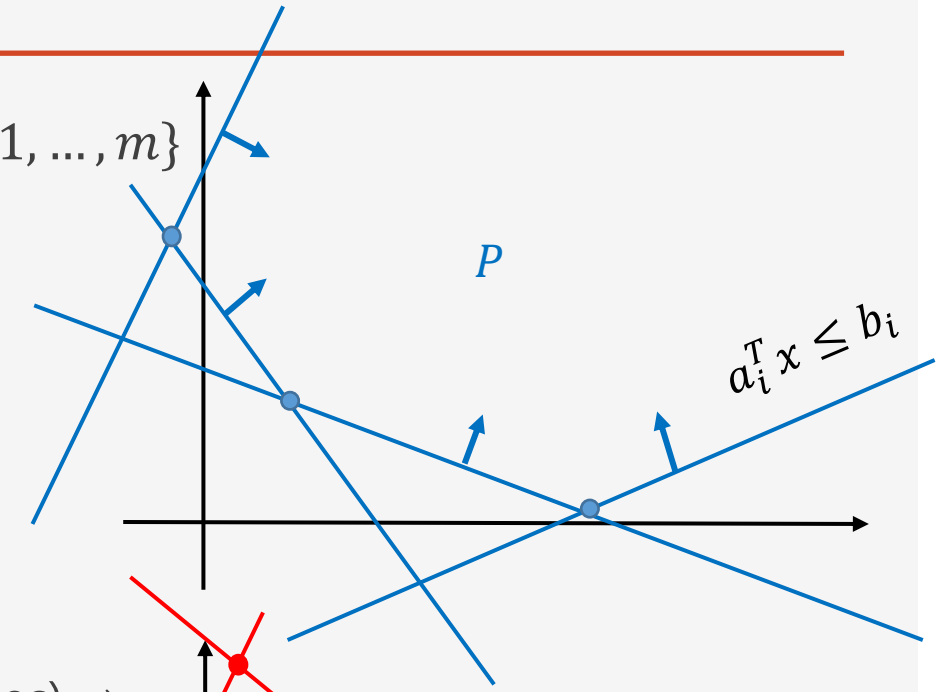
Note: If $P = I$, the Ellipsoid becomes a Ball

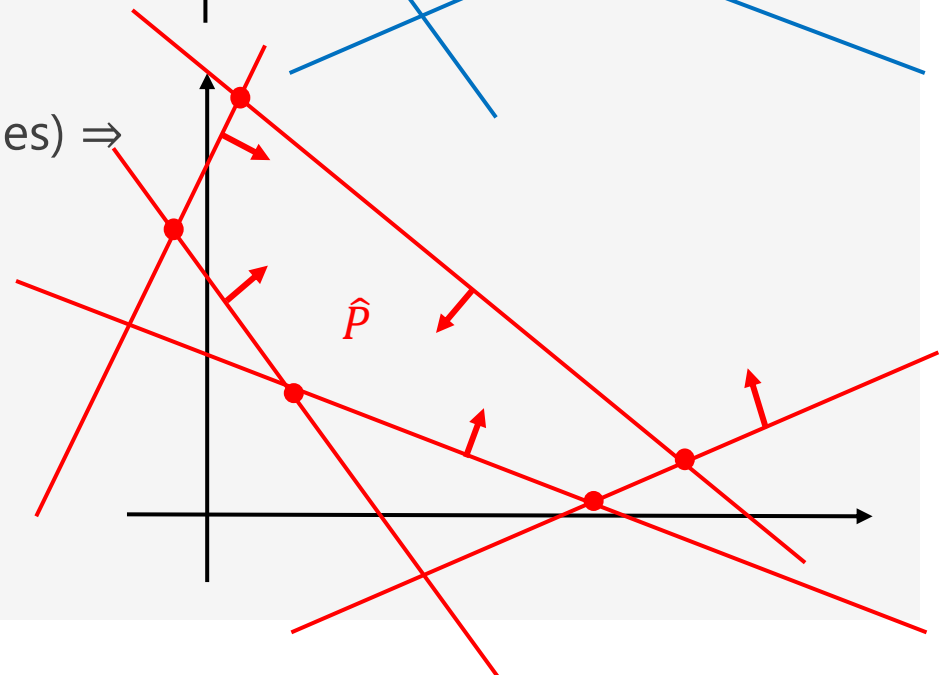
Frequently used convex sets

- **Polyhedron**: $P = \{x \in R^n \mid A^T x \leq b\} = \{x \in R^n \mid a_i^T x \leq b_i\ , i = 1, \dots, m\}$

$$A = \begin{bmatrix} -a_1^T - \\ \dots \\ -a_m^T - \end{bmatrix} \qquad A \in R^{m \times n},\ a_i \in R^n,\ b \in R^m$$

- A lot of applications in Linear Optimization
- It is an intersection of m Halfspaces ($a_i^T x \leq b_i\ ,\ i = 1, \dots, m$)
- <u>The intersection of convex sets is a convex set</u> (in a couple of slides) ⇒
  - **Polyhedrons are convex sets.**
- A polyhedron *may be an unbounded set*.
- **Polytope**: $\hat{P}$ is a *bounded* polyhedron





Convex sets consisted of Matrices

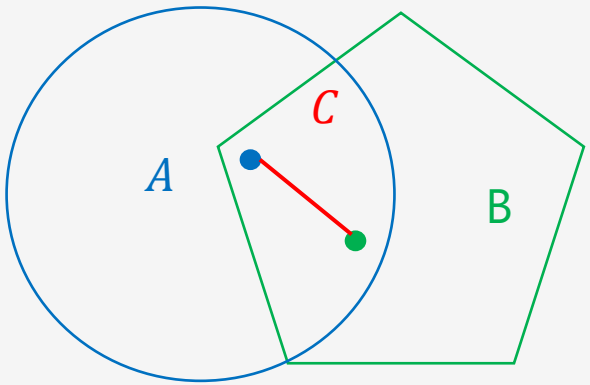
- **Theorem**: The set of (symmetric) PSD matrices is a convex set
  - $S_+^n = \{P \in R^{n\times n} | \, P \succcurlyeq 0\}$

- Proof: next slides!
- **Theorem**: The set of non-negative polynomials in n variables and of degree d is a convex set
- We will discuss more about them in the near future…
  - Semidefinite Programming & Polynomial Optimization

Preserving convexity in sets

- **Theorem**: $A, B \subseteq R^n$ convex sets $\Rightarrow C = A \cap B$ is a convex set

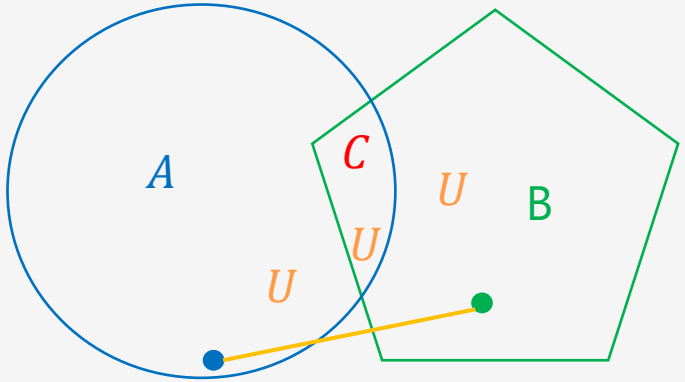
- **Proof**: as Exercise!

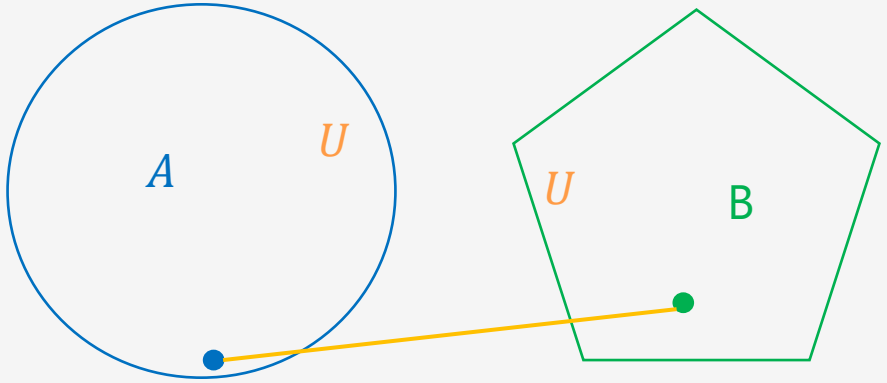


Intersection of convex sets preserves convexity

Remember the feasible region of a Convex Optimization problem??

Remark: The union $U = A \cup B$ of two convex sets A, B is not always a convex set!





Convex sets - Examples

Euclidean balls are convex sets: $B_c = \{x \in R^n \mid ||x - c|| \leq r\}$

<u>Proof</u>: take any $x, y \in B_c$ and $\lambda \in [0,1]$. Need to show: $\lambda x + (1-\lambda)y \in B_c$ or $||\lambda x + (1-\lambda)y - c|| \leq r$.

Indeed: $||\lambda x + (1-\lambda)y - c|| = ||\lambda x + (1-\lambda)y - \lambda c - (1-\lambda)c|| =$

$$= ||\lambda(x-c) + (1-\lambda)(y-c)|| \leq ||\lambda(x-c)|| + ||(1-\lambda)(y-c)|| \leq \lambda r + (1-\lambda)r = r$$

The of all PSD matrices are convex sets: $\mathcal{F} = \{A \in R^{n\times n} \mid A \succcurlyeq 0\}$

<u>Proof</u>: take any two PSD matrices $A, B \in F$ and $\lambda \in [0,1]$.

Need to show $\lambda A + (1-\lambda)B \succcurlyeq 0$

Indeed: $\forall x \in R^n,\ x^T(\lambda A + (1-\lambda)B)x = \lambda x^T A x + (1-\lambda)x^T B x \geq 0$
because $A \succcurlyeq 0$ and $B \succcurlyeq 0$.

$$S_2 = \left\{(x,y,z) \mid \begin{bmatrix} x & y \\ y & z \end{bmatrix} \succcurlyeq 0\right\}$$

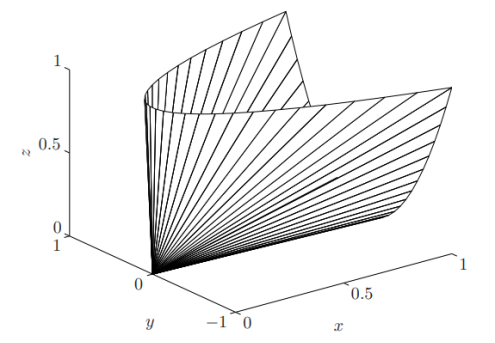


Figure 2.12 Boundary of positive semidefinite cone in $\mathbf{S}^2$.

Example 2.6 in [BV04] book

Convex functions

- **Theorem:** Affine functions $f(x) = c^T x + b$ are convex functions

- **Theorem:** Quadratic functions $f(x) = x^T Q x + c^T x + b$ are convex functions iff $Q \succcurlyeq 0$ (strictly convex iff $Q \succ 0$

- **Theorem**: Any norm is a convex function (do you remember the general definition of a norm??)

  - $L_2(x) = ||x||_2$, $L_1(x) = \left|\left|x\right|\right|_1$, $L_\infty(x) = \max\{|x_i| : i = 1,2,\dots,n\}$

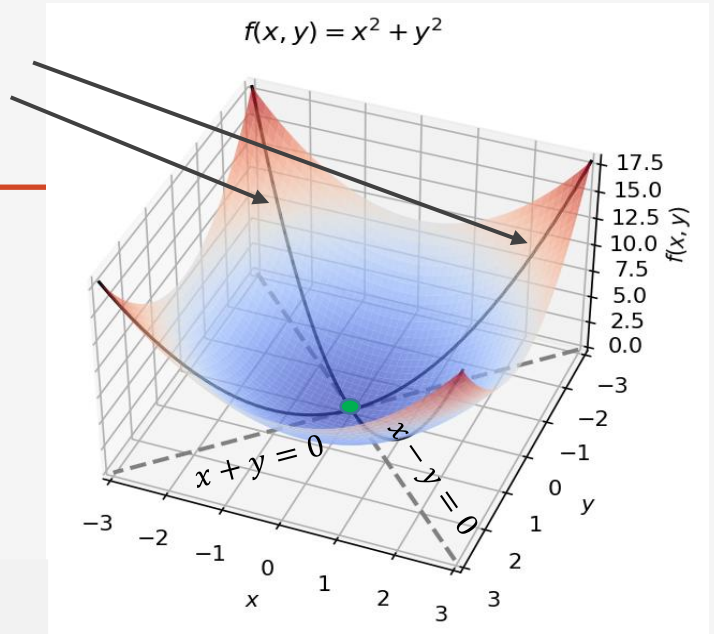
Proof: as Exercise!

- Questions:

  - 1. is an affine function concave??

  - 2. when a quadratic function is concave??

Convexity and *"Convexity Along All Lines"*

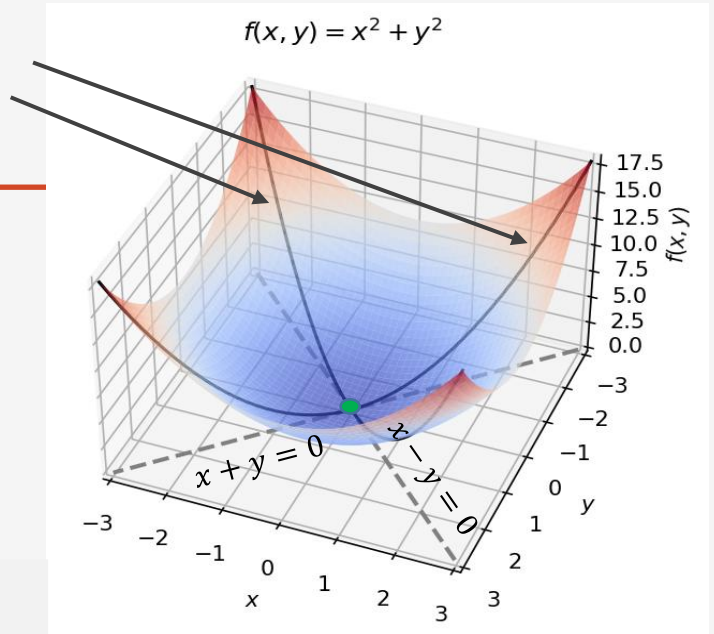


- **Theorem**: $f: R^n \to R$ is convex function $\Leftrightarrow$ the function $g: R \to R$, with $g(t) = f(x + ty)$ is convex function for all $x \in dom(f) \subseteq R^n$ and $y \in R^n$ and $t \in R$.

Proof: We will use **proof by contradiction!**

Assume that $\exists\, x \in dom(f)$ & $y \in R^n$: $g(t) = f(x + ty)$ is not convex.

Convexity and *"Convexity Along All Lines"*



- **Theorem**: $f: R^n \to R$ is convex function $\Leftrightarrow$ the function $g: R \to R$, with $g(t) = f(x + ty)$ is convex function for all $x \in dom(f) \subseteq R^n$ and $y \in R^n$ and $t \in R$.

Proof: We will use **proof by contradiction!**

Assume that $\exists\, x \in dom(f)$ & $y \in R^n$: $g(t) = f(x + ty)$ is not convex.

Then $\exists\, t_1, t_2 \in R$ & $\lambda \in [0,1]$: $g\big(\lambda t_1 + (1 - \lambda t_2)\big) > \lambda g(t_1) + (1 - \lambda) g(t_2)$ (1)

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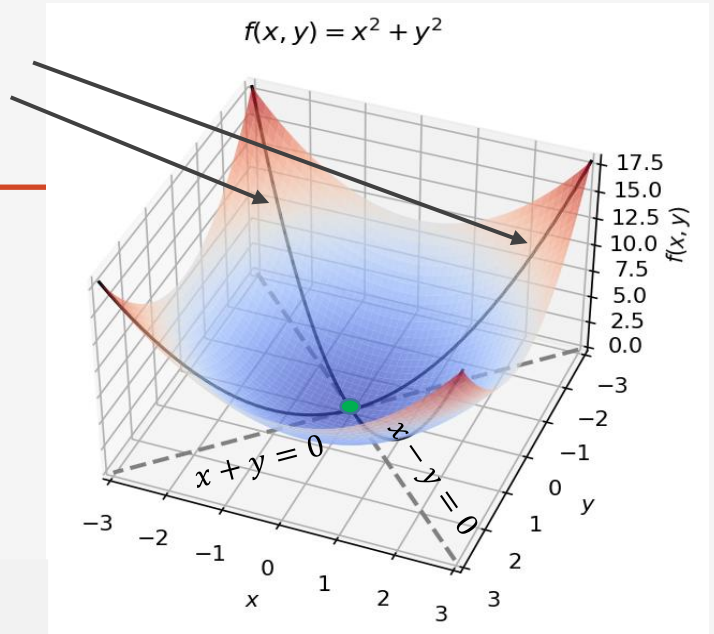

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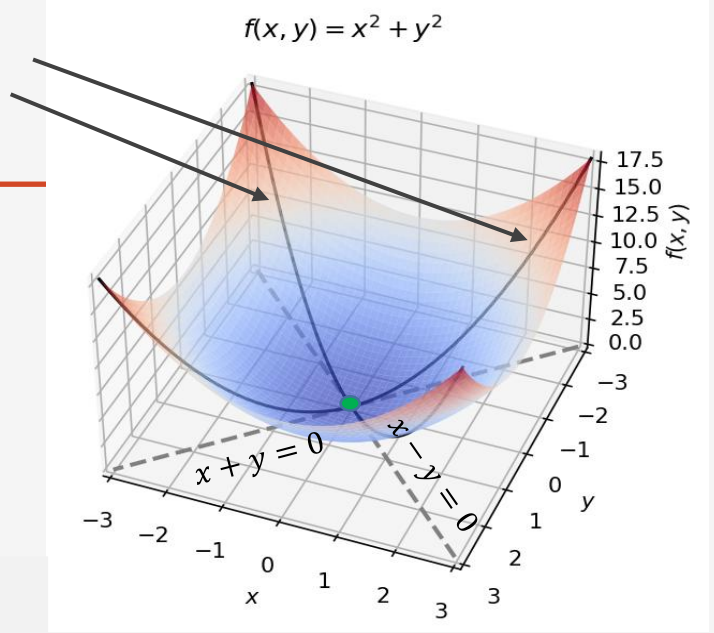
But by definition: $g\big(\lambda t_1 + (1 - \lambda) t_2)\big) = f(x + (\lambda t_1 + (1 - \lambda) t_2) y) = f(\lambda x + (1 - \lambda) x + (\lambda t_1 + (1 - \lambda) t_2) y) =$

f **is convex**

$= f(\lambda(x + t_1 y) + (1 - \lambda)(x + t_2 y))$

Convexity and *"Convexity Along All Lines"*



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$$= f(\lambda(x + t_1 y) + (1 - \lambda)(x + t_2 y)) \overset{f \text{ is convex}}{\leq} \lambda f(x + t_1 y) + (1 - \lambda) f(x + t_2 y) = \lambda g(t_1) + (1 - \lambda) g(t_2) \qquad (2)$$

As you can see **(1) and (2) contradict each other**. Hence our assumption that g is not convex was wrong. ■

Convexity and *"Convexity Along All Lines"*

- **Theorem**: $f: R^n \to R$ is convex function $\Leftrightarrow$ the function $g: R \to R$, with $g(t) = f(x + ty)$ is convex function for all $x \in dom(f) \subseteq R^n$ and $y \in R^n$ and $t \in R$.

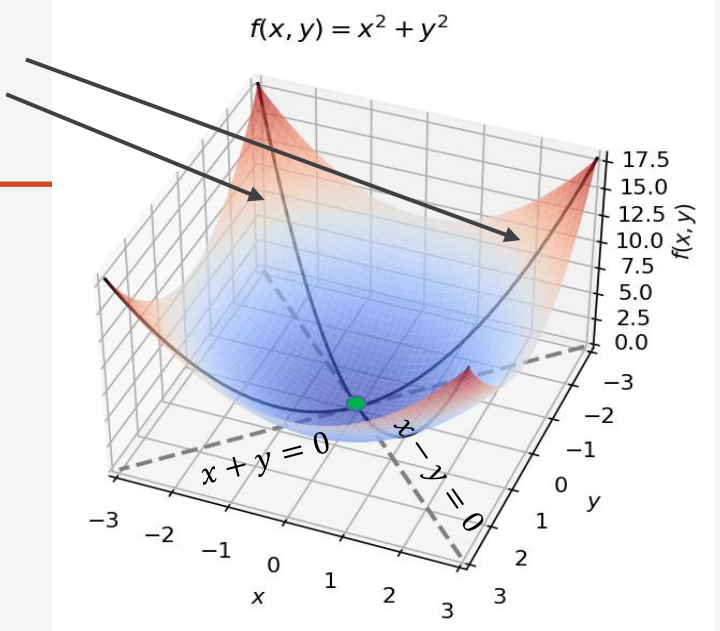
- Observations:

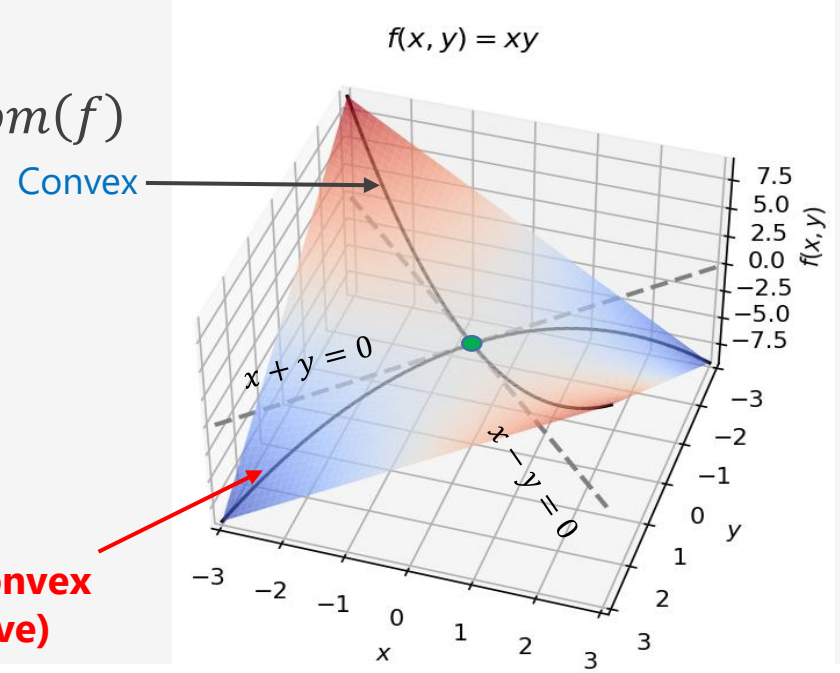
  - The domain of g is all points $t \in R$ such that $x + ty \in dom(f)$

- **Remark**:

  - If the function f is not convex, then there is a line along which the function $g(t) = f(x + ty)$ is not convex

Proof: By contradiction

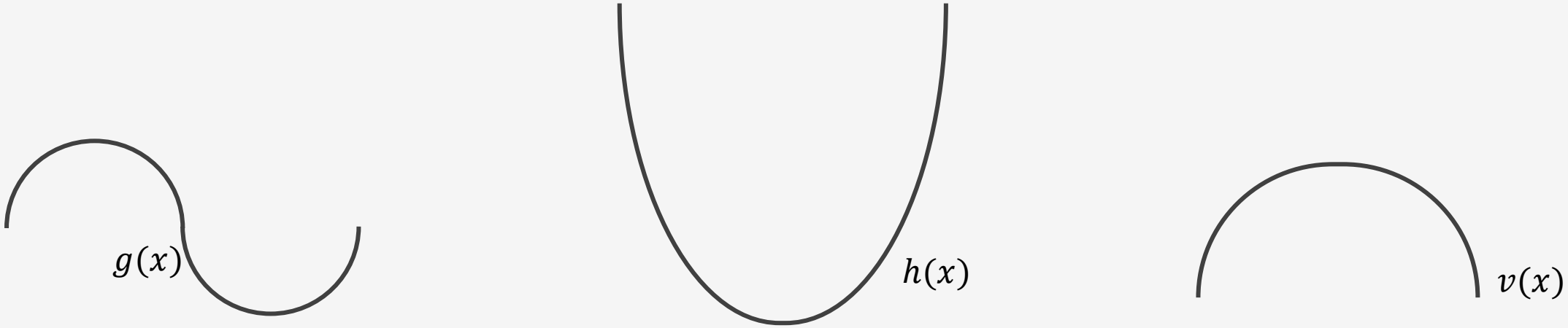




Epigraph → connecting Convex sets and Convex functions

- **Definition**: $epi(f) = \{(x,t) \mid x \in dom(f),\ f(x) \leq t\} \subseteq R^{n+1}, \quad x \in R^n, t \in R$

- **Theorem**: $f: R^n \to R$ is convex function $\Leftrightarrow$ $epi(f)$ is convex set



Epigraph is a Greek word

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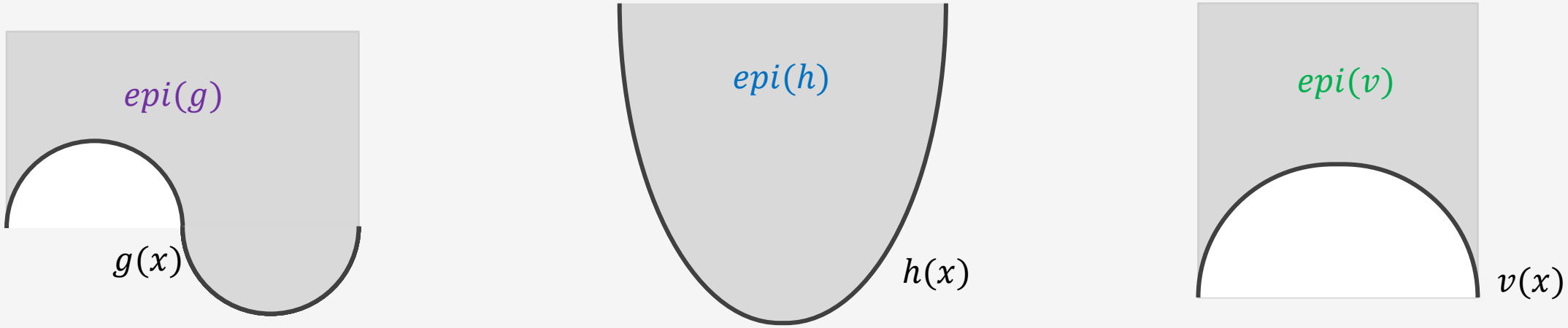

"Epi" means "Above" in Greek

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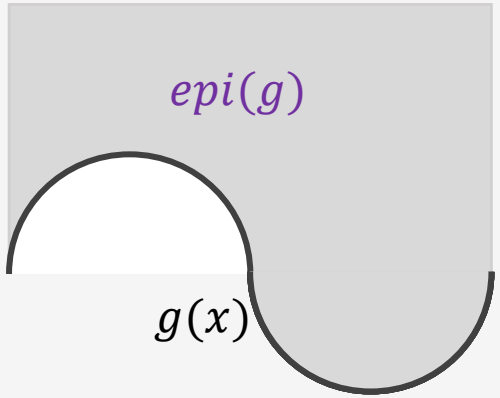
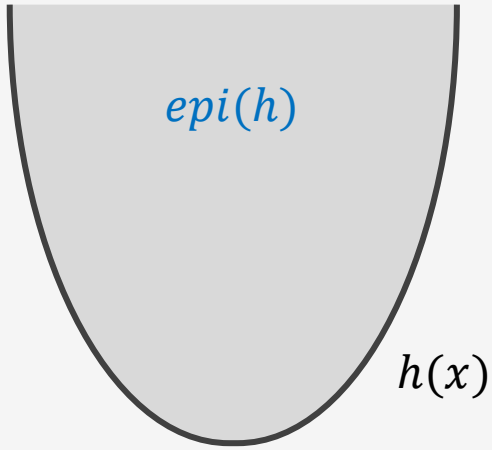
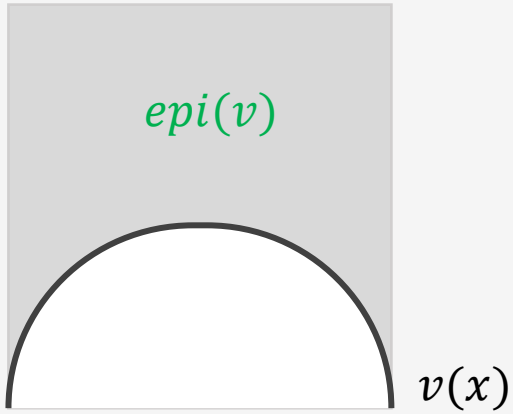


g is nonconvex function
$epi(g)$ is not convex set



h is convex function
$epi(h)$ is convex set



v is concave function
$epi(v)$ is not convex set

Epigraph → connecting Convex sets and Convex functions

- **Theorem**: $f: R^n \to R$ is convex function $\Leftrightarrow$ $epi(f)$ is convex set

- **Proof**: $\Rightarrow$ Take any $\mathrm{X} = (x, t), Y = (y, t) \in epi(f)$ and $\lambda \in [0,1]$. We **need to show**: $\lambda X + (1-\lambda)Y \in epi(f)$ or equivalently: $f(\lambda x + (1-\lambda)y) \leq t$ →(eq.1)

- Since f is convex, we have: $f(\lambda x + (1-\lambda)y) \leq \lambda f(x) + (1-\lambda)f(y)$ →(eq.2)

- Since $(x, t), (y, t) \in epi(f)$ we have: $f(x) \leq t$ and $f(y) \leq t$ →(eq.3)

- (eq.2) and (eq.3) $\Rightarrow$ $f(\lambda x + (1-\lambda)y) \leq \lambda t + (1-\lambda)t = t$

- Hence $\lambda X + (1-\lambda)Y \in epi(f)$ and therefore $epi(f)$ is convex set

Epigraph → connecting Convex sets and Convex functions

- **Proof**: *Continued from previous slide.*
- ⇐ Proof by contradiction. Assume that f is not convex. Then $\exists x, y \in dom(f)$ and $\lambda \in [0,1]$ st
  - $f(\lambda x + (1-\lambda)y) > \lambda f(x) + (1-\lambda)f(y)$ →(eq.4)
- Pick any two points such as: $\mathrm{X} = \big(x, f(x)\big) \in epi(f)$ and $\mathrm{Y} = \big(y, f(y)\big) \in epi(f)$
- We know $epi(f)$ is convex. Hence: $\mathrm{Z} = \lambda X + (1-\lambda)Y \in epi(f) \Rightarrow$

$$\Rightarrow \big(\lambda x + (1-\lambda)y, \lambda f(x) + (1-\lambda)f(y)\big) \in epi(f)$$

- From the definition of $epi(f)$ we have: $f(\lambda x + (1-\lambda)y) \leq \lambda f(x) + (1-\lambda)f(y)$ →(eq.5)
- From (eq.4) and (eq.5) we can see we arrived at a contradiction. Hence f is convex function ■

Convexity of sub-level sets (let's make some more connections)

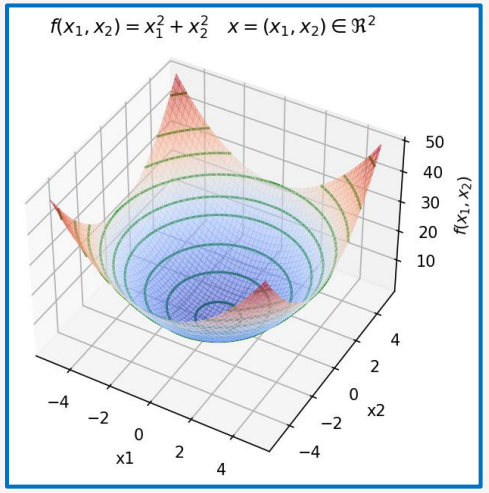
- **Theorem**: If function $f: R^n \to R$ is convex, then ($\Rightarrow$) all its sublevel sets are convex sets
- Do you remember the definition of α-sublevel set??
  - How about the α-level set??

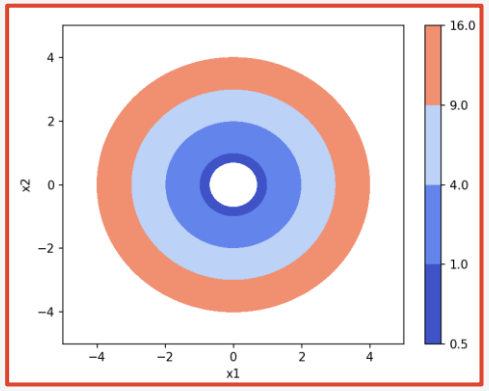
Convexity of sub-level sets

- **Theorem**: If function $f: R^n \to R$ is convex, then ($\Rightarrow$) all its sublevel sets are convex sets

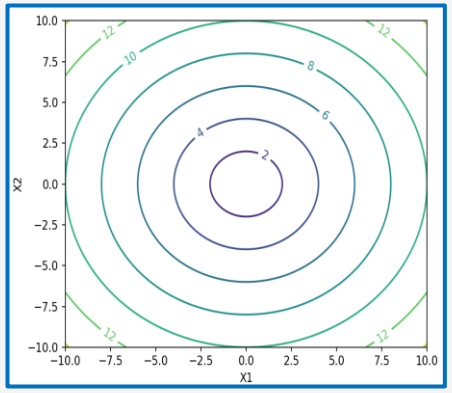
- Do you remember the definition of α-sublevel set??
  - $S_\alpha = \{x \in dom(f) \mid f(x) \leq a\}$

- **Proof**: as exercise





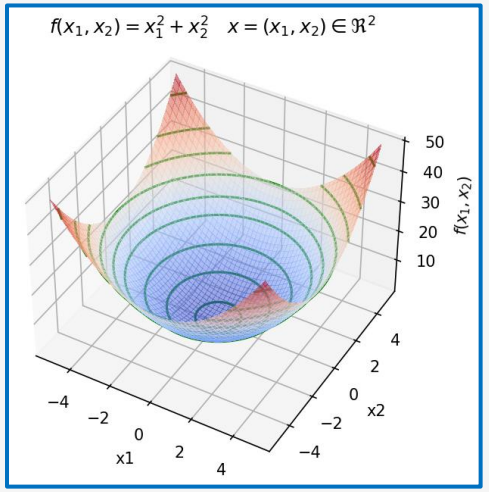
Sublevel sets



Level sets

Convexity of sub-level sets

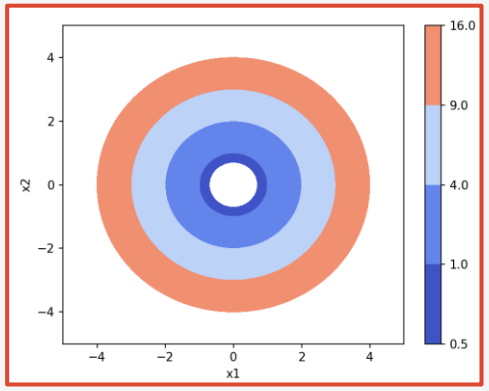
- **Theorem**: If function $f: R^n \to R$ is convex, then (⇒) all its sublevel sets are convex sets



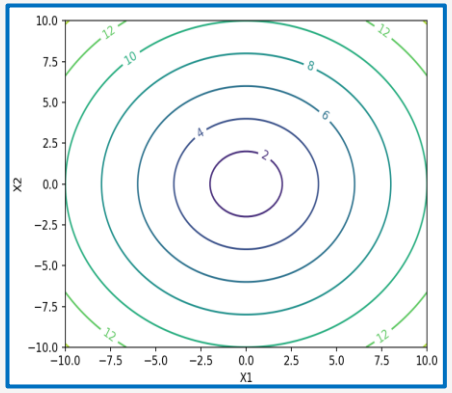
- Do you remember the definition of α-sublevel set??
  - $S_\alpha = \{x \in dom(f) \mid f(x) \leq a\}$

- **Proof**: as exercise

Remark: the converse is NOT true.
Can you think of a **counterexample**??



Sublevel sets



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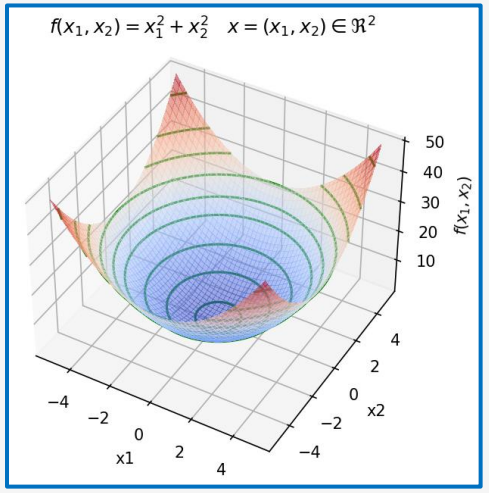

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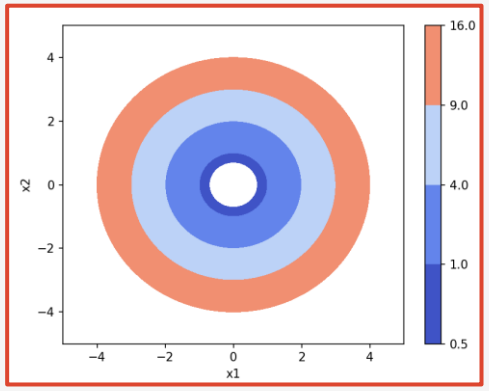
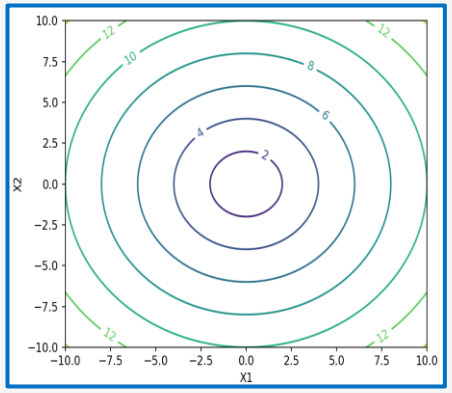
- **Proof**: as exercise

Remark: the converse is NOT true.
Can you think of a **counterexample**??
$f(x) = x^3$ (plot its sublevel sets…)



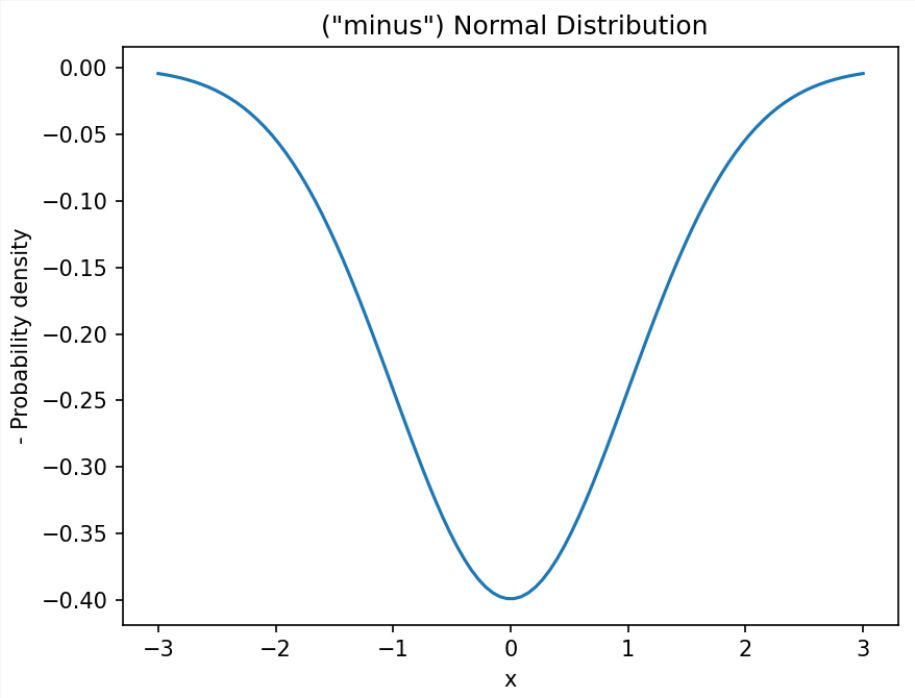
Sublevel sets



Level sets

Quasiconvex functions

- **Definition**: A function $f: R^n \to R$ is called quasiconvex if all its sublevel sets are convex



Let's find some sublevel sets

Probability Density = $f(x) =- e^{-(x-\mu)^2/(2*\sigma^2)} /(\sigma \ *\sqrt{(2\pi)})$

Quasiconvex functions

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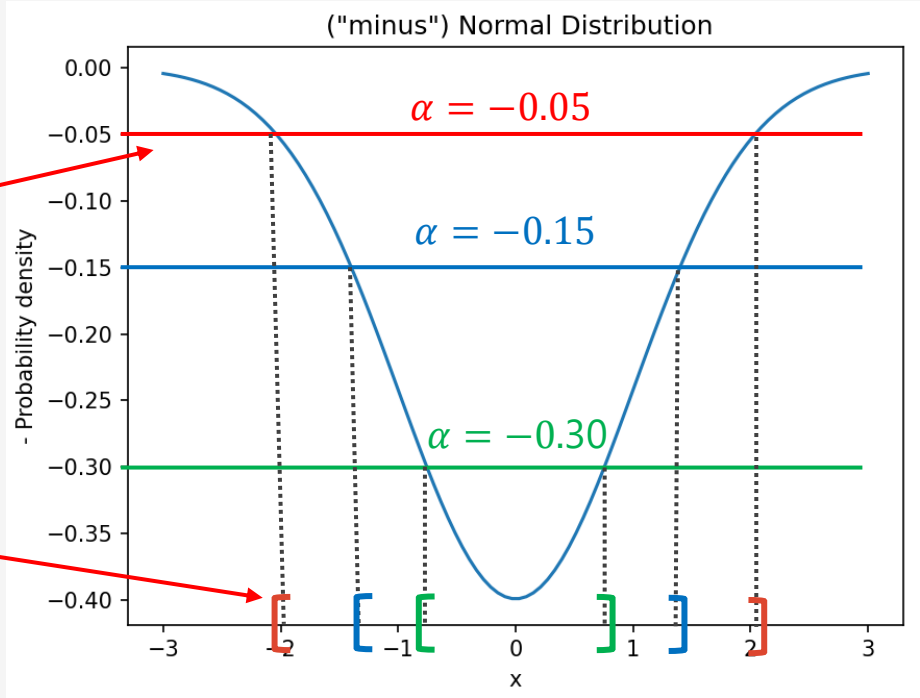
$\alpha -$ level set

$\alpha -$subevel set



Is a quasiconvex function, convex??

Probability Density = $f(x) =- e^{-(x-\mu)^2/(2*\sigma^2)} /(\sigma \ *\sqrt{(2\pi)})$